World-Model Value Depends on the Policy That Acts

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OUR POSITION

Evaluate world models together with the policies that act on their predictions.

A policy with better baseline performance can gain more from the same advice.

+11.50 mm

Change in advice gain
BC100 → BC300

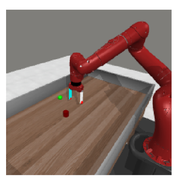
01 Predict, score, select—then let the policy act.

Task: push the red object to the green goal in simulation (Meta-World [1]). **BC:** behavior cloning from expert actions; policy input is simulator state.

Generate nine plans	Predict future features	Score each candidate	Keep or replace
1 BC plan + 8 modified plans 20 actions per plan	Frozen JEPA-WM [2] At step 10: forecast to step 30	Learned distance estimate Lower score is better	Replace the BC plan only if estimated gain exceeds m

Candidates: add/subtract four random hand-motion changes; keep the gripper unchanged. **Scorer:** distance regression + pairwise ranking from simulation.
Inputs: current/goal features + candidate actions + predicted features (WM) or current features (direct). Margin m is selected on validation data.

Same saved state



BC300 reaches step 10

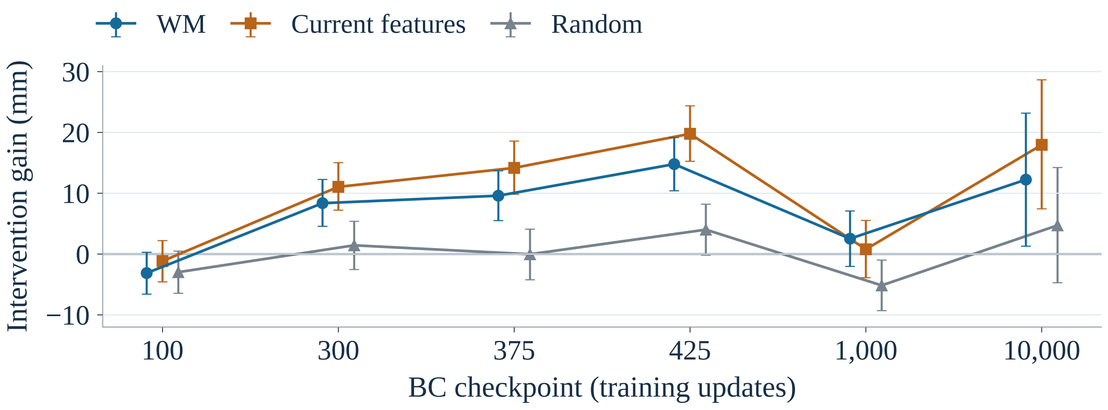
Execute five stored actions (Steps 10–15)

- Baseline BC300 actions → **Same test policy in both runs** (Replan every five actions; steps 15–100): BC100, BC300, ... → **Final distance** (Mean over steps 91–100): Baseline error
- WM-selected actions → BC100, BC300, ... → Error with advice

Advice = five selected actions. Fix the state, candidates, forecasts and both five-action sequences; vary the policy used next. BC100/300 denote 100/300 training updates.

02 Same advice, larger gain at BC300

512 shared states; one five-action intervention.
Scorer target: distance after 20 candidate actions.



Final distance and gain (mm)

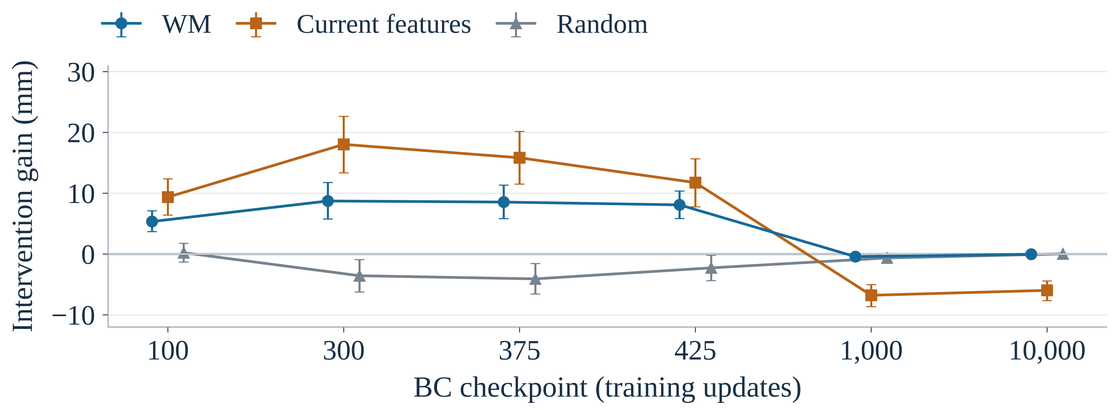
Policy	Baseline ↓	With advice ↓	Gain ↑
BC100	143.51	146.64	**−3.13**
BC300	121.08	112.71	**+8.37**
BC375	115.36	105.77	**+9.60**

BC100 → BC300: baseline improves **22.43 mm**;
advice gain rises **11.50 mm** [6.80, 16.25].

BC375 shows the same direction in means (descriptive).
All six checkpoints come from one BC training run.

03 Benefit changes across acting policies

1,024 cases; select repeatedly; baseline = BC alone.
Scorer target: final error after five actions + BC control.



World-model selection	Current-observation selection
+8.73 mm	**+18.04 mm**

At BC300, current-observation selection gains more.

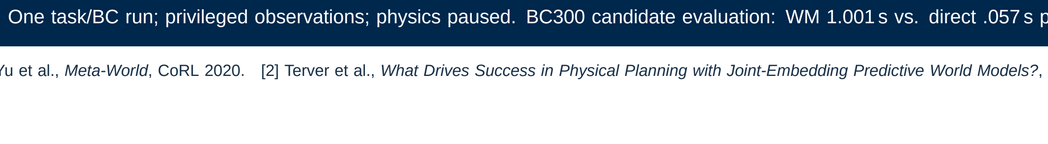
Stronger standalone policies: **BC1000 / BC10000**.
Baseline: 22.53 / 20.58 mm; WM gain: −0.43 / −0.03 mm.

BC300 success: **14.2% → 20.3% with WM (+6.2 pp)**.
Success: final distance ≤50 mm; direct scoring: 35.4%.

Gain = baseline − advice error; positive is better. Bars: paired 95% intervals. The controlled BC100 → BC300 claim passes the prespecified family-adjusted tests (15 tests; α = .010).
The two protocols differ in scoring targets, intervention schedules and baselines. WM/direct use matched capacity and supervision; margins are validated separately.
Diagnostic (post hoc; 256 separate cases): execute the plan with lowest actual 20-step distance vs. the unmodified 20-action plan. At BC1000: endpoint improves 7.79 mm; final error worsens 3.68 mm (Appendix).

04 Matching forecasts to candidate actions improves selection

Exploratory: 512 reused states; BC300; five-step predicted features. Scorers receive BC300 actions at eight probe states. Values are final-task gains.



Shuffling breaks action–forecast correspondence during both training and evaluation; all nine candidates are used.

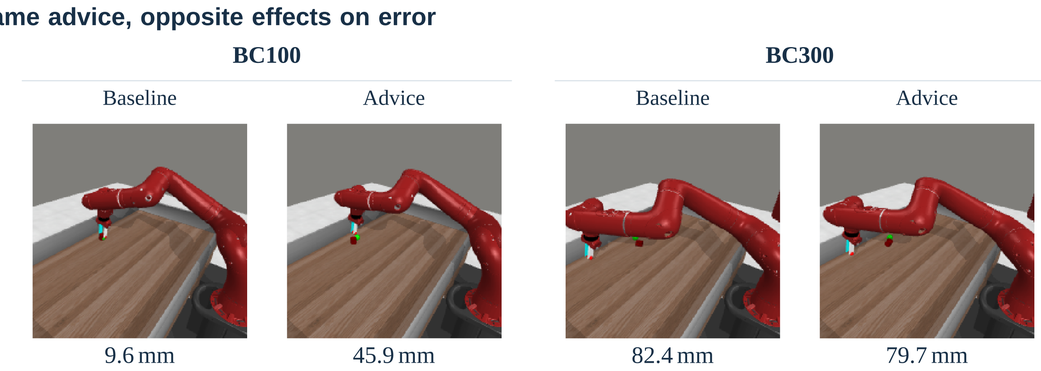
Correct pairing	Shuffled pairing	Current features
17.08 mm	**5.48 mm**	**15.13 mm**

Matched − shuffled: **+11.60 mm** [7.37, 15.90].
Matched − current: +1.95 mm [−0.96, 4.88]; superiority is unresolved.

05 Robot examples: the same actions have different effects

Exploratory cases with the same BC300 action description supplied to the scorer. Images show step 100; errors average steps 91–100.

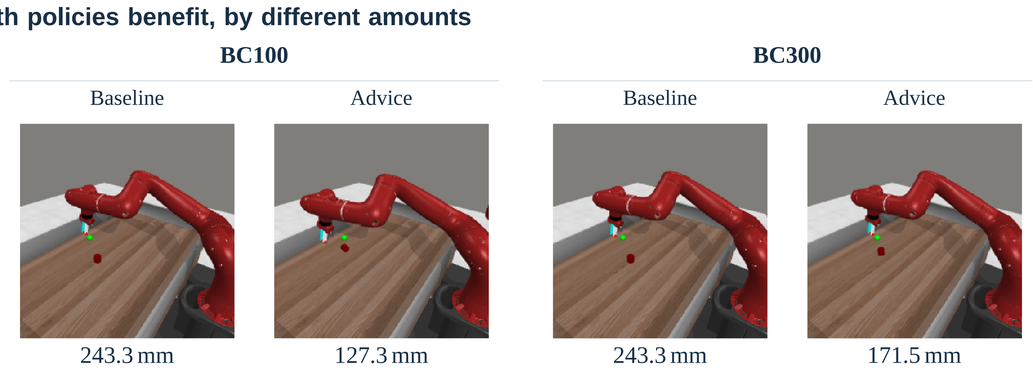
(a) Same advice, opposite effects on error



Gain: −36.3 → +2.6 mm; baseline error increases.

Cases selected after evaluation near each sign group's median. Across 512 cases: 252 increased / 177 decreased / 83 unchanged gains.

(b) Both policies benefit, by different amounts



Gain: +116.0 → +71.8 mm.

Evaluate world models with acting policies through task completion.

One task/BC run; privileged observations; physics paused. BC300 candidate evaluation: WM 1.001 s vs. direct .057 s per episode.

[1] Yu et al., *Meta-World*, CoRL 2020. [2] Terver et al., *What Drives Success in Physical Planning with Joint-Embedding Predictive World Models?*, 2025.

QR: Paper + robot videos